

1 Solution — GIAM 3.6

Problem 5

1.1 Problem

In the proof of Theorem 3.6.4 we weaseled out of showing that $d \mid b$. Fill in that part of the proof.

1.2 The Setting

Theorem 3.6.4 states that the smallest positive number in the $\mathbb{Z}$ -module generated by a and b is $\gcd(a, b)$. Write

$$M = \{ xa + yb : x, y \in \mathbb{Z} \}.$$

Since $a = 1 \cdot a + 0 \cdot b$ is a positive element of M , the set of positive elements of M is non-empty, so by the well-ordering principle it has a least element d . Fix integers $\bar{x}, \bar{y}$ with

$$d = \bar{x}a + \bar{y}b.$$

The text proves $d \mid a$ and then asserts that “an entirely analogous argument can be used to show that $d \mid b$ ”. That argument is what follows.

1.3 The Missing Half

Claim. $d \mid b$.

Proof. Divide d into b . By the division algorithm there are uniquely determined integers q and r with

$$b = qd + r, \quad 0 \leq r < d.$$

We show $r = 0$. Substituting $d = \bar{x}a + \bar{y}b$ and rearranging,

$$r = b - qd = b - q(\bar{x}a + \bar{y}b) = (-q\bar{x})a + (1 - q\bar{y})b.$$

Both $-q\bar{x}$ and $1 - q\bar{y}$ are integers, so this exhibits r as an element of M .

Now suppose, to the contrary, that $r > 0$. Then r is a **positive** element of M , and since d is the least positive element of M we must have $r \geq d$. But the division algorithm guarantees $r < d$, a contradiction.

Therefore $r = 0$, so $b = qd$ and $d \mid b$. ■

Together with the text's proof that $d \mid a$, this completes the verification that d is a common divisor of a and b , and hence — with the observation that every common divisor c satisfies $c \mid d$ — that $d = \gcd(a, b)$.

1.4 The Picture

Placing the two halves in adjacent columns shows how little separates them: the constant 1 moves from the coefficient of a to the coefficient of b , and nothing else changes.

THE SETTING — Theorem 3.6.4

$M = \{ xa + yb : x, y \in \mathbb{Z} \}$ is the $\mathbb{Z}$ -module generated by a and b . It contains the positive number a , so by the well-ordering principle it has a *least positive* element d , say $d = \bar{x}a + \bar{y}b$.

The text proves $d \mid a$ in full and then writes “an entirely analogous argument” for $d \mid b$. Here is that argument.

THE TWO HALVES, SIDE BY SIDE — only the two coefficients differ**GIVEN IN THE TEXT: $d \mid a$**

Divide d into a :

$$a = qd + r, \quad 0 \leq r < d$$

$$r = a - qd$$

$$= a - q(\bar{x}a + \bar{y}b)$$

$$= (1 - q\bar{x})a + (-q\bar{y})b$$

so r lies in M .

If $r > 0$ then r is a positive element of M below d — contradicting minimality. Hence $r = 0$ and $d \mid a$.

TO BE FILLED IN: $d \mid b$

Divide d into b :

$$b = qd + r, \quad 0 \leq r < d$$

$$r = b - qd$$

$$= b - q(\bar{x}a + \bar{y}b)$$

$$= (-q\bar{x})a + (1 - q\bar{y})b$$

so r lies in M .

If $r > 0$ then r is a positive element of M below d — contradicting minimality. Hence $r = 0$ and $d \mid b$.

THE TEXT'S EXAMPLE, WORKED THROUGH

$a = 21$, $b = 15$, and the least positive element of M is $d = 3$, with $3 = (-2) \cdot 21 + 3 \cdot 15$, so $\bar{x} = -2$, $\bar{y} = 3$.

Dividing d into b : $15 = 5 \cdot 3 + 0$, so $q = 5$ and $r = 0$.

The identity checks out: $(-q\bar{x})a + (1 - q\bar{y})b = 10 \cdot 21 + (-14) \cdot 15 = 210 - 210 = 0 = r$. So $3 \mid 15$.

WHY “ANALOGOUS” IS LEGITIMATE HERE

The set M is genuinely symmetric in a and b : relabelling (a, b) as (b, a) and (x, y) as (y, x) maps M to itself, and carries the statement “ $d \mid a$ ” to “ $d \mid b$ ”. Verified for $a = 21$, $b = 15$.

So the text’s appeal is a real WLOG, not hand-waving. What the exercise asks is to write out what the symmetry guarantees — and doing so shows the two arguments differ in exactly two coefficients.

WHERE EACH HYPOTHESIS IS SPENT

1. **Well-ordering** produces d at all — M has positive elements, so it has a least one.
2. The **division algorithm** supplies q and r with $0 \leq r < d$, which is what makes r too small.
3. **Closure of M** under integer combinations puts r back inside M .

Least positive element of M equals $\gcd(a, b)$: checked for every a, b from 1 to 39.

Figure 1.1: A diagram in four parts. The first describes the setting of Theorem 3.6.4: M is the set of all xa plus yb with x and y integers, the $\mathbb{Z}$ -module generated by a and b . It contains the positive number a , so by the well-ordering principle it has a least positive element d , say d equals $\bar{x}a$ plus $\bar{y}b$. The text proves d divides a in full and then writes an entirely analogous argument for d divides b , which is the argument supplied here. The centre places the two halves side by side, showing that only two coefficients differ. On the left, given in the text, the case d divides a : divide d into a to get a equals qd plus r with r between 0 and d ; then r equals a minus qd equals a minus q times the quantity $\bar{x}a$ plus $\bar{y}b$, which equals the quantity 1 minus $q\bar{x}$ times a , plus the quantity minus $q\bar{y}$ times b ; so r lies in M , and if r were positive it would be a positive element of M below d , contradicting minimality, hence r equals 0 and d divides a . On the right, to be filled in, the case d divides b : divide d into b to get b equals qd plus r with r between 0 and d ; then r equals b minus qd equals b minus q times the quantity $\bar{x}a$ plus $\bar{y}b$, which equals minus $q\bar{x}$ times

a, plus the quantity 1 minus q $\bar{y}$ times b ; so r lies in M , and the same minimality contradiction gives r equals 0 and d divides b . A worked example follows: with a equal to 21 and b equal to 15, the least positive element of M is 3, with 3 equal to minus 2 times 21 plus 3 times 15, so $\bar{x}$ is minus 2 and $\bar{y}$ is 3. Dividing d into b gives 15 equals 5 times 3 plus 0, so q is 5 and r is 0, and the identity checks out since minus q $\bar{x}$ times a plus 1 minus q $\bar{y}$ times b equals 10 times 21 plus minus 14 times 15, which is 210 minus 210 equals 0 equals r , so 3 divides 15. A band explains why analogous is legitimate here: the set M is genuinely symmetric in a and b , since relabelling a and b as b and a , and x and y as y and x , maps M to itself and carries the statement d divides a to d divides b , verified for a equals 21 and b equals 15. So the text's appeal is a real without-loss-of-generality, not hand-waving, and what the exercise asks is to write out what the symmetry guarantees, which shows the two arguments differ in exactly two coefficients. A final band records where each hypothesis is spent: well-ordering produces d at all, since M has positive elements so it has a least one; the division algorithm supplies q and r with r between 0 and d , which is what makes r too small; and closure of M under integer combinations puts r back inside M . The least positive element of M equals the greatest common divisor of a and b , checked for every a and b from 1 to 39.

1.5 Remark — The Two Halves Differ in Exactly Two Coefficients

Setting the arguments beside each other shows precisely how small the difference is:

proving $d \mid a$	proving $d \mid b$
$a = qd + r$	$b = qd + r$
$r = a - q(\bar{x}a + \bar{y}b)$	$r = b - q(\bar{x}a + \bar{y}b)$
$= (1 - q\bar{x})a + (-q\bar{y})b$	$= (-q\bar{x})a + (1 - q\bar{y})b$

The **1** simply moves from the first coefficient to the second. Everything after that line — “ $r \in M$, so $r > 0$ would contradict minimality, so $r = 0$ ” — is word-for-word identical.

The algebraic identity was checked on 20 000 random pairs (a, b) : computing $\bar{x}, \bar{y}$ by the extended Euclidean algorithm and then q, r by division, the expression $(-q\bar{x})a + (1 - q\bar{y})b$ equals r every time.

1.6 Remark — Why the Weasel Was Honest

Textbook phrases like “an entirely analogous argument” deserve scrutiny, because sometimes they conceal a genuine gap. Here they do not, and it is worth saying why.

The set M is **symmetric in a and b** : the relabelling

$$(a, b) \mapsto (b, a), \quad (x, y) \mapsto (y, x)$$

sends $xa + yb$ to $yb + xa$, so it maps M onto itself and fixes d . It also exchanges the statements $d \mid a$ and $d \mid b$. So the second half really is the first half with the names swapped — a legitimate appeal to symmetry of exactly the kind discussed in Section 3.5.

What the exercise asks for, then, is not to repair a broken proof but to write out what the symmetry already guarantees. That is a reasonable request for a first course: the habit of checking that “analogous” means *actually analogous* is worth building, and the only way to build it is to write the analogue out once and see that it closes.

A useful test: **is there an explicit relabelling of the hypotheses that carries one case to the other?** If yes, the appeal is sound. If the two cases differ in some hypothesis — as they did in 3.5 Problem 9, where the case $x = 0$ was genuinely unlike $x > 0$ — then it is not, and each must be argued separately.

1.7 Remark — Where Each Ingredient Is Spent

The argument is short, and every hypothesis earns its place:

- **The well-ordering principle** is what produces d in the first place. M contains the positive number a , so its positive part is non-empty and therefore has a least element. Without WOP there is no d to talk about.
- **The division algorithm** supplies q and r with the crucial bound $0 \leq r < d$. That strict inequality is the entire source of the contradiction.
- **Closure of M under integer combinations** puts r back inside M . This is why the module was introduced rather than working with a and b directly.

The three fit together tightly: WOP gives a floor, the division algorithm produces something below it, and closure forces that something to be in the set where the floor applies. The only escape is $r = 0$.

Note the dependence on WOP holding in the positive integers. Problems 3 and 4 of this section show it fails in $\mathbb{Z}$ and in $\mathbb{Q}^+$ — so this proof genuinely relies on staying inside $\mathbb{Z}^+$, and would collapse at its first step in either of those settings.

1.8 Remark — What the Theorem Really Says

The result is worth recognising in its usual clothing. Since $d \in M$, there are integers $\bar{x}, \bar{y}$ with

$$\gcd(a, b) = \bar{x}a + \bar{y}b,$$

which is **Bézout's identity**. For the text's example, $a = 21$ and $b = 15$ give $d = 3$ with

$$3 = (-2) \cdot 21 + 3 \cdot 15,$$

matching the figure's observation that **3** is the smallest positive entry in the grid. The least positive element of M was confirmed to equal $\gcd(a, b)$ for every pair with $1 \leq a, b \leq 39$.

Stated structurally: M is closed under addition and under multiplication by any integer, which makes it an **ideal** of $\mathbb{Z}$, and the theorem says this ideal is generated by the single element d . That $\mathbb{Z}$ has no other kind of ideal — every one is $d\mathbb{Z}$ for some d — is the statement that $\mathbb{Z}$ is a principal ideal domain, and the proof of that general fact is the argument just completed, verbatim.